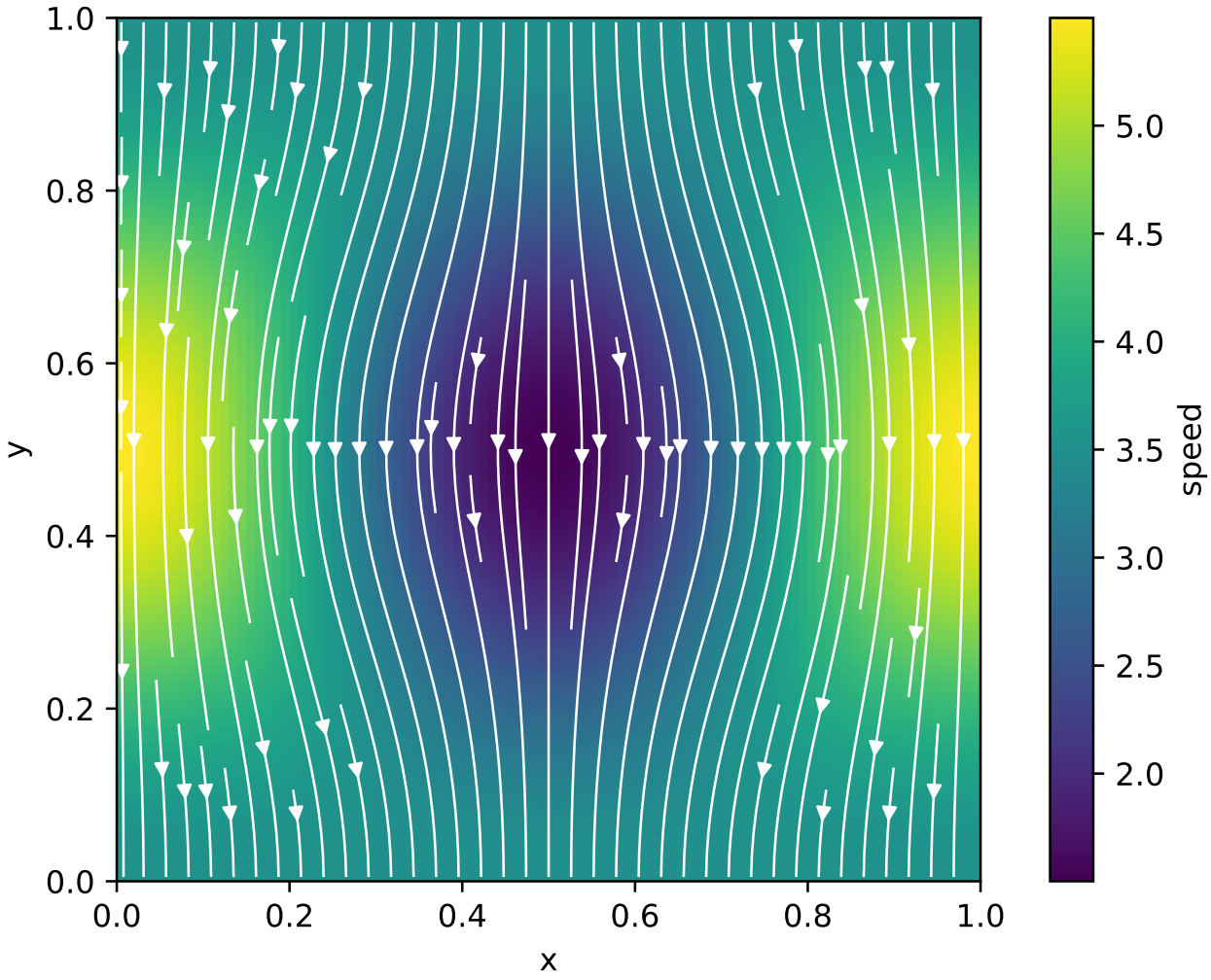
Sean Gravelle
Numerical PDEs Homework #1

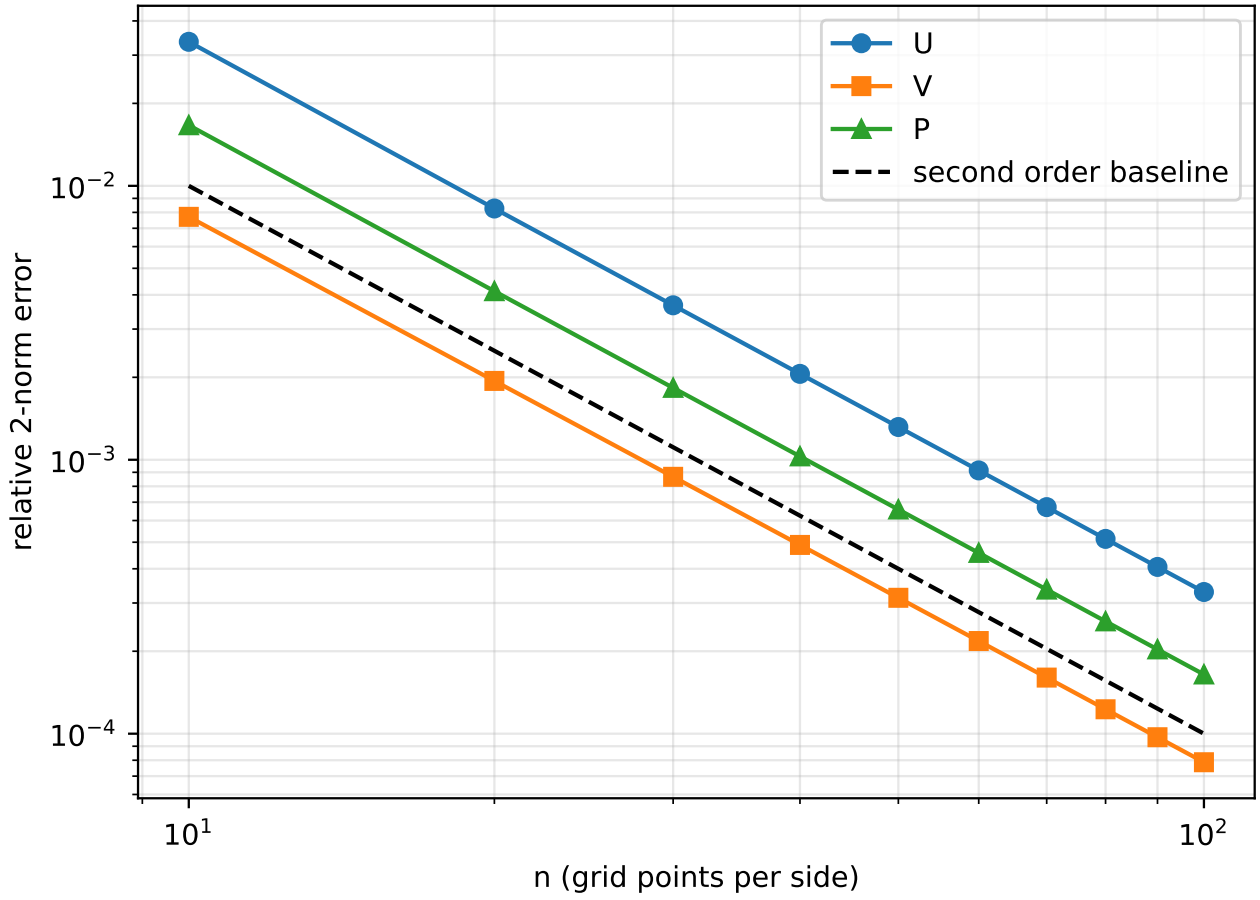
Problem 1: Solve Stokes

My solution to the Stokes equations in 2 dimensions, with the given boundary conditions, is plotted below. I used $n = 100 = Nx = Ny$ gridpoints in both x and y directions. Arrows indicate direction of the fluid's flow and color indicates the magnitude of the solution vector; that is, the speed of the fluid.



Problem 2: Error Analysis

Plotting the relative error of numerical solutions on an $n \times n$ grid gives the following errors for U, V, and P. The slopes of each show that their errors decrease proportionally to $1/n^2$ (the second order baseline plotted as a black dashed line).



AI disclosure statement: All plots were made with LLM assistance. All mathematical work and all code to produce the numerical solutions and error calculations are my own, without LLM.